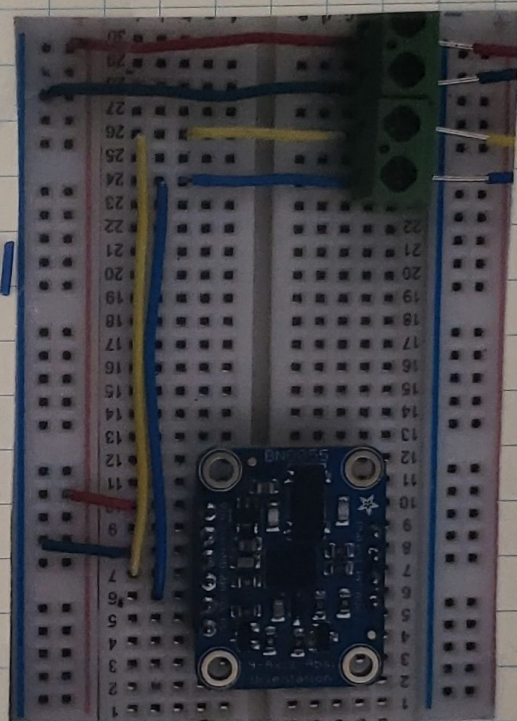
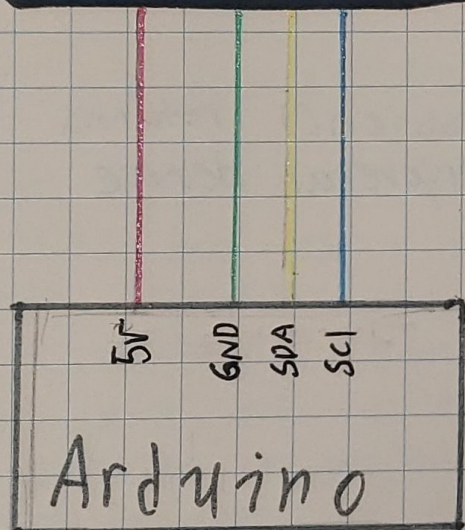
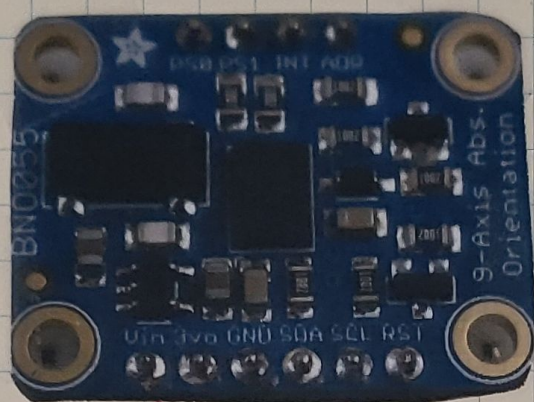


The BNO055 is an orientation sensor containing an accelerometer, gyroscope, and magnetometer

There is a library available for using the BNO, but we don't use it due to issues with measuring heading encountered by Wesley (refer to his notebooks).



Serial Clock (SCL) and Serial data (SDA) are the two signal lines in I2C.

- the controller/central drives the clock line
- the peripheral sends data to the central on the data line
- when the peripheral is receiving data, the central places the serial data bits on the data line

A device connected to the I2C bus does not drive the signal lines high or low. It can only pull these lines low using an open-drain connection.

The operating mode effects what can be done with the sensor

→ different parts of the sensor will turn off or on depending on the operating mode

Table 3-3: Operating modes overview

Operating Mode		Available sensor signals			Fusion Data	
		Accel	Mag	Gyro	Relative orientation	Absolute orientation
Non-fusion modes	CONFIGMODE	-	-	-	-	-
	ACCONLY	X	-	-	-	-
	MAGONLY	-	X	-	-	-
	GYROONLY	-	-	X	-	-
	ACCMAG	X	X	-	-	-
	ACCGYRO	X	-	X	-	-
	MAGGYRO	-	X	X	-	-
Fusion modes	AMG	X	X	X	-	-
	IMU	X	-	X	X	-
	COMPASS	X	X	-	-	X
	M4G	X	X	-	X	-
	NDOF_FMC_OFF	X	X	X	-	X
	NDOF	X	X	X	-	X

Table 3-5: operating modes selection

Parameter	Value	[Reg Addr]: Reg Value
CONFIG MODE	CONFIGMODE	[OPR_MODE]: xxx0000b
	ACCONLY	[OPR_MODE]: xxx0001b
	MAGONLY	[OPR_MODE]: xxx0010b
	GYROONLY	[OPR_MODE]: xxx0011b
	ACCMAG	[OPR_MODE]: xxx0100b
	ACCGYRO	[OPR_MODE]: xxx0101b
	MAGGYRO	[OPR_MODE]: xxx0110b
Fusion Mode	AMG	[OPR_MODE]: xxx0111b
	IMU	[OPR_MODE]: xxx1000b
	COMPASS	[OPR_MODE]: xxx1001b
	M4G	[OPR_MODE]: xxx1010b
	NDOF_FMC_OFF	[OPR_MODE]: xxx1011b
	NDOF	[OPR_MODE]: xxx1100b

register

4.3.27 EUL_DATA_X_LSB 0x1A

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r	r	r	r	r	r	r	r
Reset	0	0	0	0	0	0	0	0
Content	Heading Data <7:0>							

DATA	bits	Description
Heading Data <7:0>	<7:0>	Lower byte of heading data, read only The output units can be selected using the UNIT_SEL register and data output type can be changed by updating the Operation Mode in the OPR_MODE register, see section 3.3

4.3.28 EUL_DATA_X_MSB 0x1B

	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Access	r	r	r	r	r	r	r	r
Reset	0	0	0	0	0	0	0	0
Content	Heading Data <15:8>							

DATA	bits	Description
Heading Data <15:8>	<15:8>	Upper byte of heading data, read only The output units can be selected using the UNIT_SEL register and data output type can be changed by updating the Operation Mode in the OPR_MODE register, see section 3.3

IF we want to get Eul-DATA-X, we need
LSB and MSB, so we'll specify 0x1A and grab
two bytes so we also get 0x1B

